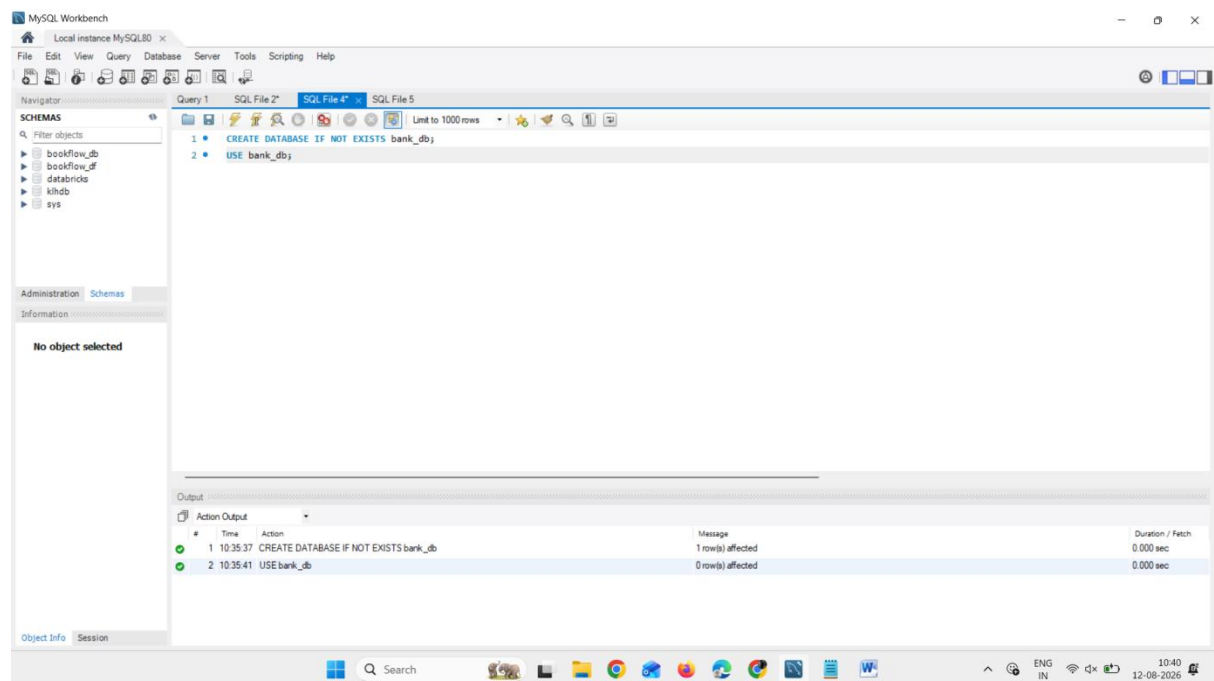


(WEEK-1)

1. Database Setup



2. Create Table

```
3 CREATE TABLE bank_transactions (  
4   txn_id INT PRIMARY KEY,  
5   customer_name VARCHAR(50),  
6   branch_name VARCHAR(50),  
7   transaction_type VARCHAR(20),  
8   amount DECIMAL(10,2),  
9   transaction_date DATE  
10 );
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
1	10:35:37	CREATE DATABASE IF NOT EXISTS bank_db	1 row(s) affected	0.000 sec
2	10:35:41	USE bank_db	0 row(s) affected	0.000 sec
3	10:42:10	CREATE DATABASE IF NOT EXISTS bank_db	1 row(s) affected. 1 warning(s): 1007 Can't create database 'bank_db'; database exists	0.000 sec
4	10:42:10	USE bank_db	0 row(s) affected	0.000 sec
5	10:42:10	CREATE TABLE bank_transactions (txn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name V...	0 row(s) affected	0.031 sec

3. Add Column

```

11 • ALTER TABLE bank_transactions
12   ADD account_no VARCHAR(20);

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
4	10:42:10	USE bank_db	0 row(s) affected	0.000 sec
5	10:42:10	CREATE TABLE bank_transactions (bn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name...	0 row(s) affected	0.031 sec
6	10:43:14	CREATE DATABASE IF NOT EXISTS bank_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_db'; database exists	0.016 sec
7	10:43:14	USE bank_db	0 row(s) affected	0.000 sec
8	10:43:14	CREATE TABLE bank_transactions (bn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name...	Error Code: 1050. Table 'bank_transactions' already exists	0.000 sec
9	10:43:20	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

4. Modify Column

```

13 • ALTER TABLE bank_transactions
14   MODIFY customer_name VARCHAR(100);

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
5	10:42:10	CREATE TABLE bank_transactions (bn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name...	0 row(s) affected	0.031 sec
6	10:43:14	CREATE DATABASE IF NOT EXISTS bank_db	1 row(s) affected, 1 warning(s): 1007 Can't create database 'bank_db'; database exists	0.016 sec
7	10:43:14	USE bank_db	0 row(s) affected	0.000 sec
8	10:43:14	CREATE TABLE bank_transactions (bn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name...	Error Code: 1050. Table 'bank_transactions' already exists	0.000 sec
9	10:43:20	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
10	10:44:26	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec

5. Rename Table

```

15 • RENAME TABLE bank_transactions TO customer_transactions;
16 • RENAME TABLE customer_transactions TO bank_transactions;

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
7	10:43:14	USE bank_db	0 row(s) affected	0.000 sec
8	10:43:14	CREATE TABLE bank_transactions (bn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name...	Error Code: 1050. Table 'bank_transactions' already exists	0.000 sec
9	10:43:20	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
10	10:44:26	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
11	10:45:12	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.016 sec
12	10:45:12	RENAME TABLE customer_transactions TO bank_transactions	0 row(s) affected	0.015 sec

6. Truncate Table

```

17 • TRUNCATE TABLE bank_transactions;

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
8	10:43:14	CREATE TABLE bank_transactions (bn_id INT PRIMARY KEY, customer_name VARCHAR(50), branch_name...	Error Code: 1050. Table 'bank_transactions' already exists	0.000 sec
9	10:43:20	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
10	10:44:26	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
11	10:45:12	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.016 sec
12	10:45:12	RENAME TABLE customer_transactions TO bank_transactions	0 row(s) affected	0.015 sec
13	10:47:15	TRUNCATE TABLE bank_transactions	0 row(s) affected	0.016 sec

7. Drop Table

```

18 • CREATE TABLE IF NOT EXISTS bank_backup (id INT);
19 • DROP TABLE bank_backup;

```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
10	10:44:26	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
11	10:45:12	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected	0.016 sec
12	10:45:12	RENAME TABLE customer_transactions TO bank_transactions	0 row(s) affected	0.015 sec
13	10:47:15	TRUNCATE TABLE bank_transactions	0 row(s) affected	0.016 sec
14	10:52:05	CREATE TABLE IF NOT EXISTS bank_backup (id INT)	0 row(s) affected	0.016 sec
15	10:52:05	DROP TABLE bank_backup	0 row(s) affected	0.016 sec

8. Insert Bulk Data

```
19 • DROP TABLE bank_backup; INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date) VALUES
20 (101, 'Ravi', 'Hyderabad', 'Deposit', 5000.00, '2024-01-05'),
21 (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000.00, '2024-01-06'),
22 (103, 'Kiran', 'Vijayawada', 'Deposit', 12000.00, '2024-01-08'),
23 (104, 'Anil', 'Vizag', 'Deposit', 8000.00, '2024-01-10'),
24 (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500.00, '2024-01-11'),
25 (106, 'Ramesh', 'Vizag', 'Deposit', 15000.00, '2024-01-12'),
26 (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000.00, '2024-01-13'),
27 (108, 'Rahul', 'Hyderabad', 'Deposit', 9000.00, '2024-01-14'),
28 (109, 'Sneha', 'Vizag', 'Withdrawal', 4000.00, '2024-01-15'),
29 (110, 'Madhu', 'Vijayawada', 'Deposit', 11000.00, '2024-01-16');
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓	12 10:45:12	RENAME TABLE customer_transactions TO bank_transactions	0 row(s) affected	0.015 sec
✓	13 10:47:15	TRUNCATE TABLE bank_transactions	0 row(s) affected	0.016 sec
✓	14 10:52:05	CREATE TABLE IF NOT EXISTS bank_backup (id INT)	0 row(s) affected	0.016 sec
✓	15 10:52:05	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
✗	16 11:16:56	DROP TABLE bank_backup	Error Code: 1051. Unknown table 'bank_db.bank_backup'	0.000 sec
✓	17 11:17:01	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

9. Insert Single Record

```
30 • INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
31 VALUES (111, 'Venu', 'Vizag', 'Deposit', 7000.00, '2024-01-18');
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓	13 10:47:15	TRUNCATE TABLE bank_transactions	0 row(s) affected	0.016 sec
✓	14 10:52:05	CREATE TABLE IF NOT EXISTS bank_backup (id INT)	0 row(s) affected	0.016 sec
✓	15 10:52:05	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
✗	16 11:16:56	DROP TABLE bank_backup	Error Code: 1051. Unknown table 'bank_db.bank_backup'	0.000 sec
✓	17 11:17:01	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
✓	18 11:18:19	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...	1 row(s) affected	0.000 sec

10. Update Data

```
32 • UPDATE bank_transactions
33 SET amount = 5000.00
34 WHERE txn_id = 105;
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓	14 10:52:05	CREATE TABLE IF NOT EXISTS bank_backup (id INT)	0 row(s) affected	0.016 sec
✓	15 10:52:05	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
✗	16 11:16:56	DROP TABLE bank_backup	Error Code: 1051. Unknown table 'bank_db.bank_backup'	0.000 sec
✓	17 11:17:01	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
✓	18 11:18:19	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...	1 row(s) affected	0.000 sec
✓	19 11:19:06	UPDATE bank_transactions SET amount = 5000.00 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

11. Delete Data

```
35 • DELETE FROM bank_transactions
36 WHERE txn_id = 111;
```

Output				
Action Output				
#	Time	Action	Message	Duration / Fetch
✓	15 10:52:05	DROP TABLE bank_backup	0 row(s) affected	0.016 sec
✗	16 11:16:56	DROP TABLE bank_backup	Error Code: 1051. Unknown table 'bank_db.bank_backup'	0.000 sec
✓	17 11:17:01	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
✓	18 11:18:19	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...	1 row(s) affected	0.000 sec
✓	19 11:19:06	UPDATE bank_transactions SET amount = 5000.00 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
✓	20 11:19:53	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.000 sec

12. Select All Records

```
37 • SELECT * FROM bank_transactions;
```

Result Grid

Filter Rows:

Edit

Export/Import:

Wrap Cell Contents:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
104	Anil	Vizag	Deposit	8000.00	2024-01-10	10000
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11	10000
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	10000
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13	10000
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	10000
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15	10000
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	10000

transactions 1 x

Apply

Revert

Output

Action Output

#

Time

Action

Message

Duration / Fetch

✖

16 11:16:56

DROP TABLE bank_backup

Error Code: 1051. Unknown table 'bank_db.bank_backup'

0.000 sec

✔

17 11:17:01

INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...

10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

0.000 sec

✔

18 11:18:19

INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction...

1 row(s) affected

0.000 sec

✔

19 11:19:06

UPDATE bank_transactions SET amount = 5000.00 WHERE txn_id = 105

1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0

0.000 sec

✔

20 11:19:53

DELETE FROM bank_transactions WHERE txn_id = 111

1 row(s) affected

0.000 sec

✔

21 11:20:34

SELECT * FROM bank_transactions LIMIT 0, 1000

10 row(s) returned

0.000 sec / 0.000 sec

13. Select With Where Filter

```
38 • SELECT * FROM bank_transactions
39 WHERE transaction_type = 'Deposit';
```

The screenshot shows the SQL Studio interface. The top pane displays a query: `SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit';`. The bottom pane shows the 'Result Grid' with 7 rows of data. The columns are: `txn_id`, `customer_name`, `branch_name`, `transaction_type`, `amount`, `transaction_date`, and `account_no`. The data rows are:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	10000
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	10000
104	Anil	Vizag	Deposit	8000.00	2024-01-10	10000
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	10000
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	10000
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	10000

The 'Action Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
17	11:17:01	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (101, 'Ravi', 'Hyderabad', 'Deposit', 5000.00, '2024-01-05', 10000)	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
18	11:18:19	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (103, 'Kiran', 'Vijayawada', 'Deposit', 12000.00, '2024-01-08', 10000)	1 row(s) affected	0.000 sec
19	11:19:06	UPDATE bank_transactions SET amount = 5000.00 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
20	11:19:53	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.000 sec
21	11:20:34	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
22	11:21:51	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec

14. Select With Order By

```
40 • SELECT * FROM bank_transactions
41 ORDER BY amount DESC;
```

The screenshot shows the SQL Studio interface. The top pane displays a query: `SELECT * FROM bank_transactions ORDER BY amount DESC;`. The bottom pane shows the 'Result Grid' with 7 rows of data. The columns are: `txn_id`, `customer_name`, `branch_name`, `transaction_type`, `amount`, `transaction_date`, and `account_no`. The data rows are:

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12	10000
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08	10000
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16	10000
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14	10000
104	Anil	Vizag	Deposit	8000.00	2024-01-10	10000
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05	10000
105	Priva	Hyderabad	Withdrawal	5000.00	2024-01-11	10000

The 'Action Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
18	11:18:19	INSERT INTO bank_transactions (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date, account_no) VALUES (101, 'Ravi', 'Hyderabad', 'Deposit', 5000.00, '2024-01-05', 10000)	1 row(s) affected	0.000 sec
19	11:19:06	UPDATE bank_transactions SET amount = 5000.00 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
20	11:19:53	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.000 sec
21	11:20:34	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
22	11:21:51	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
23	11:22:33	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

15. Create Users

```
42 • CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Password1231';
43 • CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Password1231';
```

The screenshot shows the SQL Studio interface. The top pane displays two queries: `CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Password1231';` and `CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Password1231';`. The bottom pane shows the 'Action Output' log:

#	Time	Action	Message	Duration / Fetch
19	11:19:06	UPDATE bank_transactions SET amount = 5000.00 WHERE txn_id = 105	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
20	11:19:53	DELETE FROM bank_transactions WHERE txn_id = 111	1 row(s) affected	0.000 sec
21	11:20:34	SELECT * FROM bank_transactions LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
22	11:21:51	SELECT * FROM bank_transactions WHERE transaction_type = 'Deposit' LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
23	11:22:33	SELECT * FROM bank_transactions ORDER BY amount DESC LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
24	11:23:25	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Password1231'	0 row(s) affected	0.031 sec

16. GRANT SELECT

```
44 • GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost';
```

The screenshot shows the SQL Studio interface. The top pane displays a query: `GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost';`. The bottom pane shows the 'Action Output' log:

#	Time	Action	Message	Duration / Fetch
25	11:27:34	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
26	11:27:45	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
27	11:29:36	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Password1231'	0 row(s) affected. 1 warning(s): 3163 Authorization ID 'BranchManager'@'localhost' already exists.	0.000 sec
28	11:29:53	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
29	11:32:43	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Password1231'	0 row(s) affected	0.047 sec
30	11:32:51	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec

17. Grant All Permissions

45	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost';
----	---

#	Time	Action	Message	Duration / Fetch
26	11:27:45	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
27	11:29:36	CREATE USER IF NOT EXISTS 'BranchManager'@'localhost' IDENTIFIED BY 'Password123!'	0 row(s) affected. 1 warning(s): 3163 Authorization ID 'BranchManager'@'localhost' already exists.	0.000 sec
28	11:29:53	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
29	11:32:43	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Password123!'	0 row(s) affected	0.047 sec
30	11:32:51	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
31	11:33:47	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.032 sec

18. Revoke Select Permission

46	REVOKE SELECT ON bank_transactions FROM 'Auditor1'@'localhost';
----	---

#	Time	Action	Message	Duration / Fetch
28	11:29:53	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	Error Code: 1410. You are not allowed to create a user with GRANT	0.000 sec
29	11:32:43	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Password123!'	0 row(s) affected	0.047 sec
30	11:32:51	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
31	11:33:47	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.032 sec
32	11:34:25	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
33	11:34:33	REVOKE SELECT ON bank_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec

19. Revoke All Permissions

47	REVOKE ALL PRIVILEGES, GRANT OPTION FROM 'BranchManager'@'localhost';
----	---

#	Time	Action	Message	Duration / Fetch
29	11:32:43	CREATE USER IF NOT EXISTS 'Auditor1'@'localhost' IDENTIFIED BY 'Password123!'	0 row(s) affected	0.047 sec
30	11:32:51	GRANT SELECT ON bank_transactions TO 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
31	11:33:47	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.032 sec
32	11:34:25	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
33	11:34:33	REVOKE SELECT ON bank_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
34	11:35:17	REVOKE ALL PRIVILEGES, GRANT OPTION FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec

20. Disable Autocommit

48	SET autocommit = 0;
----	---------------------

#	Time	Action	Message	Duration / Fetch
31	11:33:47	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.032 sec
32	11:34:25	GRANT ALL PRIVILEGES ON bank_transactions TO 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
33	11:34:33	REVOKE SELECT ON bank_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected	0.000 sec
34	11:35:17	REVOKE ALL PRIVILEGES, GRANT OPTION FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
35	11:35:49	REVOKE ALL PRIVILEGES, GRANT OPTION FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
36	11:35:51	SET autocommit = 0	0 row(s) affected	0.000 sec

21. Commit Transaction

49	UPDATE bank_transactions
50	SET amount = 6000.00
51	WHERE txn_id = 101;
52	COMMIT;

#	Time	Action	Message	Duration / Fetch
34	11:35:17	REVOKE ALL PRIVILEGES, GRANT OPTION FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
35	11:35:49	REVOKE ALL PRIVILEGES, GRANT OPTION FROM 'BranchManager'@'localhost'	0 row(s) affected	0.000 sec
36	11:35:51	SET autocommit = 0	0 row(s) affected	0.000 sec
37	11:36:30	COMMIT	0 row(s) affected	0.000 sec
38	11:36:38	COMMIT	0 row(s) affected	0.000 sec
39	11:36:41	UPDATE bank_transactions SET amount = 6000.00 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

22. Savepoint and Rollback

52	COMMIT;
53	SAVEPOINT Before_Update;
54	UPDATE bank_transactions
55	SET amount = 99999.00
56	WHERE txn_id = 102;
57	ROLLBACK TO Before_Update;

#	Time	Action	Message	Duration / Fetch
37	11:36:30	COMMIT	0 row(s) affected	0.000 sec
38	11:36:38	COMMIT	0 row(s) affected	0.000 sec
39	11:36:41	UPDATE bank_transactions SET amount = 6000.00 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
40	11:37:30	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
41	11:37:30	UPDATE bank_transactions SET amount = 99999.00 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
42	11:37:30	ROLLBACK TO Before_Update	0 row(s) affected	0.015 sec

23. Full Transaction Workflow

```

58 • UPDATE bank_transactions
59   SET amount = 7000.00
60   WHERE txn_id = 101;
61 • SAVEPOINT SP1;
62 • UPDATE bank_transactions
63   SET amount = 9000.00
64   WHERE txn_id = 102;
65 • ROLLBACK TO SP1;
66 • COMMIT;

```

Output

#	Time	Action	Message	Duration / Fetch
✓ 42	11:37:30	ROLLBACK TO Before_Update	0 row(s) affected	0.015 sec
✓ 43	11:38:27	UPDATE bank_transactions SET amount = 7000.00 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
✓ 44	11:38:27	SAVEPOINT SP1	0 row(s) affected	0.000 sec
✓ 45	11:38:27	UPDATE bank_transactions SET amount = 9000.00 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
✓ 46	11:38:27	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
✓ 47	11:38:27	COMMIT	0 row(s) affected	0.032 sec

24. Enable Autocommit

```

67 • SET autocommit = 1;

```

Output

#	Time	Action	Message	Duration / Fetch
✓ 43	11:38:27	UPDATE bank_transactions SET amount = 7000.00 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
✓ 44	11:38:27	SAVEPOINT SP1	0 row(s) affected	0.000 sec
✓ 45	11:38:27	UPDATE bank_transactions SET amount = 9000.00 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
✓ 46	11:38:27	ROLLBACK TO SP1	0 row(s) affected	0.000 sec
✓ 47	11:38:27	COMMIT	0 row(s) affected	0.032 sec
✓ 48	11:39:04	SET autocommit = 1	0 row(s) affected	0.000 sec